LINUX USER MANAGEMENT

Cybersecurity Internship Practical Report

Environment	JSLinux (browser-based Linux terminal)
Operating System	Alpine Linux v3.23
Practical	User creation, group management and directory permissions

1. Objective

To learn basic Linux account administration by creating a user account, creating a group, adding the user to the group, and configuring directory ownership and access permissions.

2. Tools and Environment

• JSLinux browser-based terminal
• Alpine Linux v3.23
• Shell terminal with root access
No separate Linux installation was required for this exercise.

3. Commands and Procedure

Create a user: `adduser -D student1`
Created the account without setting a password during creation.

Verify the user: `id student1`
Displayed the account UID, primary GID and group membership.

Create a group: `addgroup interns`
Created a group named interns.

Add user to group: `addgroup student1 interns`
Added student1 to the interns group.

Create a directory: `mkdir -p /tmp/intern-lab`
Created a directory for permission testing.

Set ownership: `chown student1:interns /tmp/intern-lab`
Set student1 as owner and interns as group owner.

Set permissions: `chmod 750 /tmp/intern-lab`
Owner: read/write/execute; group: read/execute; others: no access.

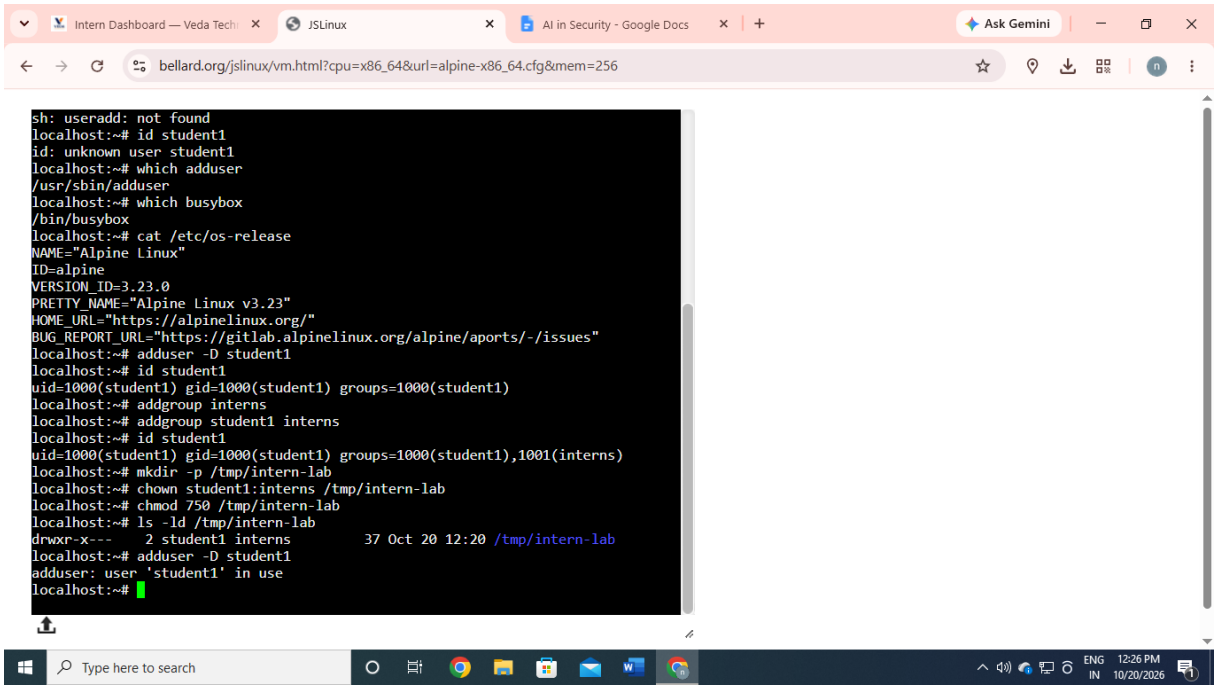
Verify permissions: `ls -ld /tmp/intern-lab`
Displayed the final directory permissions and ownership.

4. Results

The terminal output confirms that student1 was created with UID 1000. The account was added to the interns group (shown as group ID 1001). The directory /tmp/intern-lab was assigned to student1:interns and permission mode 750 was applied. The permission string shown in the screenshot is drwxr-x---.

5. Screenshot Evidence

The following screenshot records the commands and visible output from the practical, including account creation, group membership, directory ownership and permissions. The repeated adduser command at the end reports that the user is already in use, which confirms the account already exists.



6. Interview Questions and Answers

Q1. What is a Linux group?
A Linux group is a collection of user accounts used to manage access to files and resources.

Q2. Why should users be separated?
Separate accounts help assign individual permissions, improve accountability and limit unnecessary access.

Q3. Why should root access be restricted?
Root has extensive administrative privileges. Restricting its use helps prevent accidental changes and reduces security risks.

7. Conclusion

This practical demonstrated basic Linux user and group management and the use of file-system permissions. It also showed how ownership and permission settings can limit access to a directory according to the principle of least privilege.

Practical completed in JSLinux / Alpine Linux. Results are documented from the attached terminal screenshot.